

Executive Summary

Python and SQL Project

(ITSM Service Desk Operations Analytics Dataset)

1. Project Overview

This project was completed to understand and analyse IT Service Management (ITSM) data. The work combines PostgreSQL and Python to prepare the data, check its quality, perform analysis, and identify useful business observations from ticket and SLA information.

The project uses ticket data and SLA log data for the period **April to July 2021**. The two datasets were connected and then merged in Jupyter Notebook using Pandas.

2. Data Used

Dataset	Records	Purpose
Ticket	180,000	Ticket and customer-related information
Ticket SLA Log	360,000	SLA target, actual time and SLA status

3. Data Preparation

After merging, the final dataset contained **360,000 rows and 29 columns**. The increase in rows is expected because each ticket has two SLA records: **First Response** and **Resolution**.

There were no duplicate rows. The **61,446 missing Customer Satisfaction Rating values** were kept as missing because they represent cases where a rating was not provided.

No invalid satisfaction ratings were found, and the IQR-based check did not identify numerical outliers.

Two useful columns were also created during preparation:

- sla_variance_minutes
- sla_performance_percentage

These columns helped make the SLA analysis easier to understand.

4. Main Findings

Finding	Observation
SLA performance	310,006 SLA records were breached, while 49,994 were met.
Resolution SLA	All 180,000 Resolution SLA records were marked as breached.
First Response SLA	49,994 records were met and 130,006 were breached.
Customer satisfaction	Average ratings were very similar for breached and met records.
Resolution code	Average actual times were very similar across resolution codes.

5. Business Understanding

The analysis shows that SLA performance is the main area that needs attention in this dataset. The Resolution SLA is particularly important because every Resolution SLA record is shown as breached.

This suggests that resolution-related processing should be investigated further.

The First Response results are comparatively better, although a large number of records are still breached. This means response time should also continue to be monitored.

Customer satisfaction did not show a large difference between breached and met SLA records in this analysis. Therefore, SLA status alone should not be treated as a direct explanation for customer satisfaction.

6. What I Learned

As a fresher, this project helped me understand how a data project moves from raw database tables to analysis.

I worked with **PostgreSQL** for the database side and **Python, Pandas and Matplotlib** for data preparation and analysis.

I also learned that checking the actual data is important before making business conclusions.

Some categorical fields in this dataset had only one value, so those comparisons were not useful for finding real patterns.

7. Conclusion

Overall, the project gave a practical understanding of **ITSM Service Desk Operations Analytics Dataset** data analysis.

The most important observation is the high number of SLA breaches, especially for **Resolution SLA**. The analysis can help an IT support team focus on resolution time, SLA variance and response performance.

This project also shows the importance of data quality and data variety. Further analysis would be more useful with a dataset containing more variation across ticket categories, priorities, statuses and channels.

THANKYOU!!!!
